DS & AI Lab Project (BSDA4001)

TerraWatch: Intelligent Satellite Image Change Detection and Analysis

MILESTONE 2: DATASET PREPARATION

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BS in Data Science and Applications

GROUP 7

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1. Introduction

TerraWatch analyses two images of the same location to detect visible building and road changes, describe them in natural language and check whether the descriptions are supported by the images. Milestone 2 establishes the data foundation: identified sources, verified image-label joins, exploratory analysis, consistent target formats and reproducible training, validation and test records [1].

The prepared primary package contains **9,544 image pairs**, each with a categorical change mask and five reference captions. The preparation is supported by an audit of all **10,077 source pairs** and a loader check of every retained record. Dataset readiness is distinguished from trained-model accuracy.

2. Dataset Identification

Dataset	Purpose and selection	Source and disposition
LEVIR-MCI / LEVIR-CC [2,3]	Primary paired imagery, building/road masks and five captions on the same records. Supports detection and description without joining unrelated collections.	Author-linked Hugging Face release. LEVIR-CC supplies the captions; it is not an independent dataset added to the training total.
SECOND-CC [4,5]	Transfer evaluation with different imagery and semantic change annotations. Tests whether behaviour observed on LEVIR transfers to another source.	Author-linked Zenodo release, CC BY 4.0. Kept separate from primary training and from its augmented copies.
WHU-CDC [6]	Building-change transfer candidate examined in the earlier experiments.	The inspected captioned mirror has no stated derivative licence. Excluded from the governed training package; historical screening is kept separate.

LEVIR-CD and S2Looking were considered for detection-only work. LEVIR-MCI was selected for the shared package because it provides both masks and captions. Supplementary datasets are not pooled into a claim of universal robustness.

3. Dataset Description and Schema

The primary release contains 10,077 pairs and 50,385 captions. Each pair consists of an earlier RGB image, a later RGB image, a mask and five separate descriptions. All input images are **256 x 256 pixels**. The source describes nominal 0.5 m imagery from Texas, USA [2,3]. Per-tile dates and geographic coordinates are absent from the acquired record schema; they are stored as null, not estimated.

Record group	Representation	Meaning
Identity and inputs	Dataset/split/filename key; two RGB PNGs	One ordered temporal pair; input shape 2 x 256 x 256 x 3
Detection target	Categorical PNG; 256 x 256 after lossless channel reduction	0 = background, 128 = road change, 255 = building change
Description target	Five caption strings and source IDs in JSON	Alternative reference descriptions, not five additional image pairs
Provenance	Revision, archive paths, SHA-256 hashes and exclusion ledger	Reproducible joins, source verification and split membership

For example, retained record **train_000010** links the matching A/B images, a mask containing both change classes and five captions. One reference reads: "some buildings appear at the bottom of the scene." The complete record example is included in the data dictionary.

4. Data Governance

Area	Recorded decision	Application
Source and rights	Retain author attribution and source-specific terms. The LEVIR-MCI card does not provide a separate blanket data licence; a code licence is not treated as an imagery licence. SECOND-CC is CC BY 4.0 [2,4].	Research workflow; raw imagery remains local. The submission includes code, identifiers, hashes, aggregate findings and two attributed illustrative source pairs, without asserting blanket commercial or redistribution rights.
Privacy and ethics	No owner identities or personal identifiers are required by the input schema.	Descriptions concern visible physical change; ownership, legality and causes are not inferred from imagery. No claim of complete privacy anonymity is made.
Quality and provenance	Preserve original files; record conflicts and exclusion reasons. Pin the source revision and hash the acquired archive.	Labels are not silently corrected. The strict joint-task package excludes incompatible supervision while retaining the original release for traceability.
Bias and reproducibility	Record geography, restricted mask classes, caption repetition and exact processing settings.	Findings apply to the inspected release and prepared cohort. Nominal source resolution does not establish per-tile measurement accuracy.

5. Exploratory Data Analysis

The audit decoded **40,308 image/label files**: A, B, categorical label and colour label for every source pair. There were **no missing joins, invalid image dimensions, empty captions or duplicate caption keys**. All masks use the expected class values. The stored grayscale label has three identical channels; the colour copy maps background to black, road change to yellow and building change to red. These representations agree pixel by pixel.

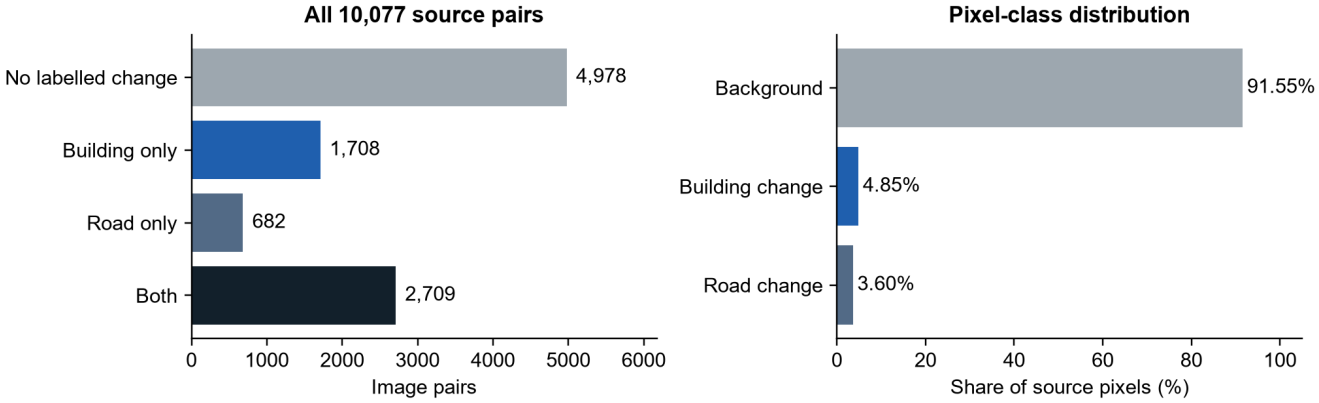


Figure 1. Complete-release statistics, calculated from all 10,077 pairs. Pair classes and pixel shares use different denominators.

Finding	Audit result	Interpretation
Mask-level balance	4,978 unchanged; 1,708 building only; 682 road only; 2,709 both	Report class-specific and unchanged-case behaviour.
Pixel imbalance	91.55% background; 4.85% building; 3.60% road	Overall pixel accuracy alone can hide missed changes.
Small changes	408 of 5,099 changed masks contain under 1% changed pixels	Preserve spatial detail and retain genuine small regions.
Mask/caption flag conflicts	511 of 10,077 pairs (5.07%)	Exclude from the strict joint-task cohort without declaring either annotation automatically correct.
Caption repetition	All 5,039 caption-flagged unchanged pairs reuse one five-sentence set	Text similarity alone can reward generic no-change descriptions.

5.1 Comparative EDA and sample annotations

A full-release comparison of **10,077 LEVIR-MCI pairs** and **6,041 original SECOND-CC pairs** shows distinct change-area and illumination distributions. Annotated change occupies **8.45%** of MCI pixels and **16.64%** of SECOND-CC pixels; the latter includes a broader semantic label vocabulary. Median temporal luminance differences are **-26.67** and **-0.61**, respectively. These findings support source-specific reporting and preservation of each dataset's label meaning. The examples below show ordered inputs, source masks and one of the five source captions for each pair [2-5].

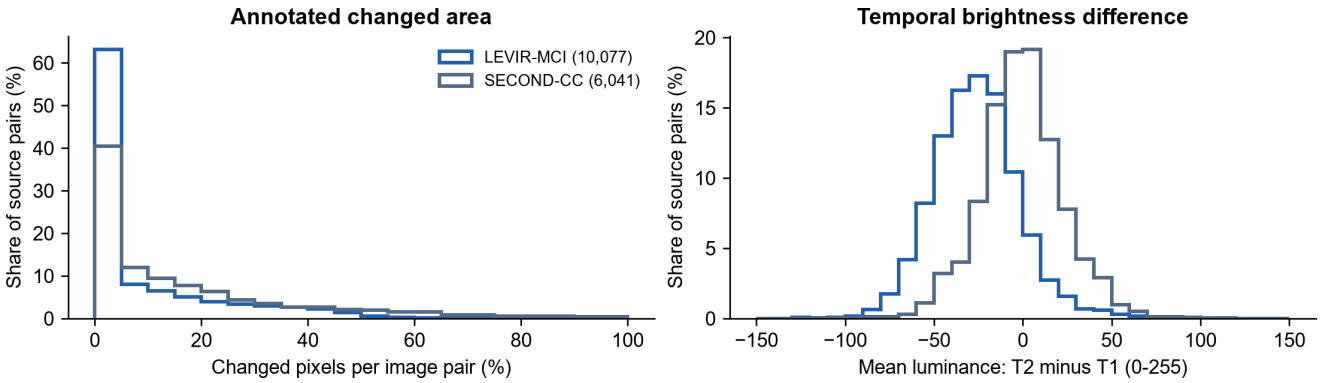


Figure 2. Distributions are normalised within each complete source release. SECOND-CC excludes the supplied augmented copies. Brightness difference is an image diagnostic, not a physical-change label.



train_000014 - Source reference: "many buildings appear alongside the road ."



train_000013 - Source reference: "a road appears in the forest ."

Figure 3. Building-change and road-change examples from the retained training partition. Display colours: white = background, red = building change, yellow = road change. Captions are supplied source references, not model predictions. Identifiers retain the original filenames; imagery and annotations are attributed to LEVIR-MCI/LEVIR-CC [2,3].

5.2 Image variation, location and reference consistency

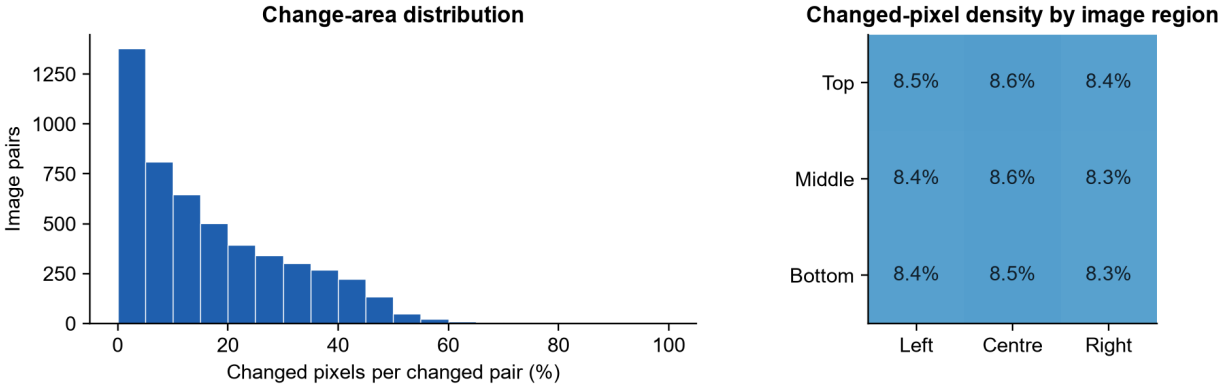


Figure 4. Change-area distribution among changed source pairs and changed-pixel density across image regions. Top/left refer to image coordinates, not geographic north/west.

Among changed masks, the median changed area is **12.60%**. Changes occur across the image, rather than only at its centre. Brightness differences also vary substantially: the median T2-minus-T1 luminance difference is **-26.67** on a 0-255 scale. Such differences are not themselves proof of physical change or poor registration.

The analysis flags 195 brightness-difference outliers using the 1.5-IQR rule. These are retained unless a separate integrity rule excludes them: extreme values can represent valid imagery. Pixel-difference, change-area, caption-length distributions and descriptive correlations are documented in the accompanying EDA record. Correlations are not treated as evidence of causation.

A lexical check found no pair mixing standard no-change phrases with change descriptions among its five references. This checks coarse change/no-change agreement, not complete agreement on object counts, direction or location. Background in a mask means **no annotated building/road change**, not absence of every possible visual change; vegetation descriptions can therefore fall outside mask coverage.

The earlier caption evaluator used a **50-pixel class-presence threshold**. This is an evaluator setting, not a dataset requirement: source building-positive pairs fall from 4,417 at one pixel to 4,356 at 50 pixels. The prepared masks retain the original small regions. Connected components are not treated as building-instance counts.

6. Data Preprocessing

- **Validate and join:** decode the paired inputs and both label forms; verify shapes, non-empty captions, class values and unique record keys. Store SHA-256 hashes of file bytes and decoded pixels. Hashing verifies identity; it is not encryption. No required input is imputed.
- **Resolve supervision conflicts:** quarantine records whose mask and caption change flags disagree, or whose repeated imagery has inconsistent targets. Preserve the source annotations and each reason in an exclusion ledger.
- **Standardise targets:** verify identical channels before reducing the grayscale label. Map values 0/128/255 to class indices 0/1/2 for background/road/building. No interpolation is applied to the native masks.
- **Prepare images:** retain RGB uint8 images for vision-language processors. The detector loader emits contiguous float32 tensors in [0,1], with temporal order preserved. No histogram matching, learned correction or automatic registration is applied.
- **Prepare text:** retain all five trimmed reference strings separately. Pair-level grouping is preserved when using multiple text targets. Checkpoint-specific tokenisation and image normalisation belong to the model processor; no fabricated merged caption is introduced.

7. Dataset Integration

A canonical **dataset / official split / filename** key links each primary pair to its detection target, five caption references and evaluation provenance.

Verified preparation trace. Record `train_000010` demonstrates the shared data contract using source annotations [2,3].

Stage	Trace for train_000010
Paired inputs	Matched earlier/later RGB images, with temporal order preserved
Detection target	Supplied road/building mask, converted to class indices 0/1/2
Caption claim	Source reference: "some buildings appear at the bottom of the scene"; object = building, action = appeared, exact quantity = unspecified, location = bottom
Visual evidence	Original T1/T2 images support inspection of the bottom region with full-frame context
Verification record	Claim fields and image provenance support the four-state human audit below

This is a source-annotation and loader trace. In model inference, masks and captions are generated from the image pair; source targets remain separate evaluation references.

SECOND-CC supplies a separate **100-pair original-validation probe** with 500 captions. Building targets derive from semantic change maps. It and WHU-CDC are excluded from primary training; source-split overlap is recorded separately.

8. Data Augmentation

Zero offline augmented samples are materialised. Native pairs and annotations are preserved. The detector training contract permits online photometric jitter and paired image-mask scaling; categorical masks use nearest-neighbour interpolation. Captioning preserves reference geometry: flips, rotations, crops and time reversal require corresponding caption validation. Constructed claim fixtures are software checks, excluded from dataset counts. Online transforms do not increase the reported dataset size.

9. Dataset Splitting and Leakage Prevention

Official split membership is preserved; records are excluded rather than moved across partitions. SHA-256 checks confirm **16 repeated image-pair groups**, including **five spanning training and test**. One copy is retained with priority test, then validation, then training. Groups with conflicting mask targets are quarantined. A cross-split perceptual-hash screen finds the same five exact pair matches and no additional candidates at the documented threshold; visual similarity is not asserted to prove scene identity.

Partition	Source pairs	Prepared pairs
Training	6,815	6,444
Validation	1,333	1,242
Test	1,929	1,858
Total	**10,077**	**9,544**

The resulting ratio is **67.52% / 13.01% / 19.47%**. All 533 exclusions have recorded reasons; reason counts can overlap. Parent-scene coordinates are unavailable, so these checks establish detected content separation, not geographic independence. Test imagery was inspected for data quality during this audit, without model fitting or checkpoint selection; it is not described as unopened.

10. Final Prepared Dataset

The final strict primary package contains **9,544 pairs, 19,088 RGB inputs, 9,544 categorical targets and 47,720 reference captions**. Its mask classes are **4,743 unchanged, 1,611 building only, 497 road only and 2,693 both**. Background, building and road occupy **91.24%, 5.11% and 3.65%** of retained pixels respectively. These figures describe the cleaned cohort, not the original release or a balanced sampling assumption.

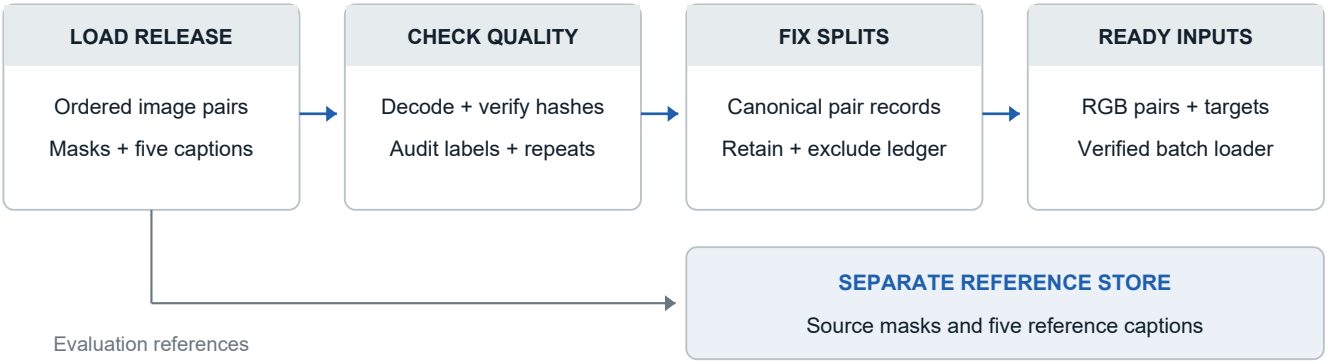


Figure 5. The preparation links source files, quality decisions and fixed partitions to a verified loader. Evaluation references remain separate from image-only prediction records.

All retained records passed hash, decoding and reference checks. Detector batches have image shape **B x 2 x 3 x 256 x 256** and target shape **B x 256 x 256**. Verified image-only inference records exclude targets and captions. Training uses the supplied loader without repeating cleaning, joins or splits.

Human factual verification protocol. Two independent raters label each atomic claim **SUPPORTED** (visually established), **CONTRADICTED** (opposed by evidence), **UNSUPPORTED** (adequate imagery without supporting evidence), or **UNVERIFIABLE** (insufficient visual information). Raw agreement and unweighted Cohen's kappa precede adjudication. A third rater resolves disagreements using zoomed T1/T2 regions and full-frame context, recording the decision and reason. Unresolved ambiguity remains UNVERIFIABLE. Source captions retain their role as references.

11. Challenges and Limitations

Duplicate imagery, inconsistent supervision and label-vocabulary differences required explicit preparation rules. The strict cohort resolves identified conflicts without declaring source annotations infallible. Background imbalance, repeated captions and limited geographic/class coverage remain dataset limitations; preparation checks do not establish generalisation. The curated cohort is distinct from the original benchmark.

12. Deliverables Produced

Deliverable	Contents
Prepared records	Separate train/validation/test manifests; image-only inference records; local source archive and reference store
Quality evidence	Full per-pair profile, exact duplicate groups, candidate checks, exclusion ledger, summary statistics and loader validation
Reproducible code	Source retrieval, full audit/preparation, dataset loader, figure generation and report builder
Documentation	Detailed EDA, data dictionary, source/usage register and preprocessing guidance

Raw data, prepared records, evidence and report outputs are separated. Versioned scripts and metadata reproduce the handover from the pinned release.

13. Summary

The audited primary data, fixed partitions and verified loader provide a reproducible handover for detection and captioning training. Data readiness, model accuracy and human factual verification remain distinct evaluation objectives.

References

1. IIT Madras DS & AI Lab. *Milestone 2: Dataset Preparation*. https://bsc-iitm.github.io/dsai-lab-project/milestones.html#milestone-2

2. Liu, C. et al. (2024). *Change-Agent: Toward Interactive Comprehensive Remote Sensing Change Interpretation and Analysis*. IEEE TGRS. Author code and LEVIR-MCI release: https://github.com/Chen-Yang-Liu/Change-Agent ; https://huggingface.co/datasets/lcybuaa/LEVIR-MCI

3. Liu, C. et al. (2022). *Remote Sensing Image Change Captioning With Dual-Branch Transformers*. IEEE TGRS. Author dataset: https://github.com/Chen-Yang-Liu/LEVIR-CC-Dataset

4. Karaca, A. C. et al. (2025). *Robust Change Captioning in Remote Sensing: SECOND-CC Dataset and MModalCC Framework*. IEEE JSTARS. https://doi.org/10.1109/JSTARS.2025.3600613 ; author-linked dataset: https://doi.org/10.5281/zenodo.16937571

5. Yang, K. et al. (2022). *Asymmetric Siamese Networks for Semantic Change Detection*. IEEE TGRS. SECOND label conventions: https://captain-whu.github.io/SCD/

6. Ji, S., Wei, S. & Lu, M. (2019). *Fully Convolutional Networks for Multisource Building Extraction From an Open Aerial and Satellite Imagery Data Set*. IEEE TGRS. WHU imagery: https://gpcv.whu.edu.cn/data/building_dataset.html ; captioned mirror inspected: https://huggingface.co/datasets/hygge10111/RS-CDC